Chapter 2

Literature review

2.1 Biological Context

Mycobacteria are a genus of bacteria with an unusually thick, waxy cell wall built around long-chain mycolic acids, which is the main reason they grow slowly and resist many standard antibiotics (Brennan, 2003). The most clinically important species, *Mycobacterium tuberculosis* (Mtb), is both slow-growing, taking three to four weeks to form a colony, and pathogenic, requiring a biosafety level 3 (BSL-3) facility to handle (Sparks et al.), which makes it very difficult to study using live, single-cell imaging. *Mycobacterium smegmatis* is used instead as a fast-growing, non-pathogenic model organism, though it is not a perfect substitute: Mtb grows and divides differently at the single-cell level, and much more slowly, doubling roughly every eighteen hours compared to three to five hours for *M. smegmatis* (Chung et al., 2024; Sparks et al.). Slow-growing pathogens therefore need much longer imaging runs to capture meaningful growth events, which is part of the motivation for building an incubator, such as the one in this project, that can hold stable conditions over extended periods.

Bulk culture measurements average away important biological detail. Even within a clonal population, individual mycobacterial cells vary in growth rate, division timing and antibiotic susceptibility, partly due to their unusual, unipolar growth pattern (Davis and Isberg, 2016; Aldridge et al., 2012). Cells also organise into larger structures: when grown without detergent, mycobacteria align into bundles known as cords, a behaviour linked to virulence that is also seen in nonpathogenic species (Julián et al., 2010). Both of these behaviours, cell-to-cell variation and cording, are only visible through live imaging at the single-cell and microcolony scale.

A common method for this kind of imaging is the agar-pad technique, where cells are sandwiched between a coverslip and a pad of agar and imaged from below on an inverted microscope (Joyce et al., 2011). This method can support long imaging runs, but the sample chamber is open to the environment rather than sealed, so the bacteria are directly and continuously affected by the surrounding conditions. Existing studies have focused on single cells or on fixed, mature colony structures, but there is little published work on long-duration, live imaging of a microcolony as it actually forms (Chung et al., 2024; Aldridge et al., 2012; Davis and Isberg, 2016; Julián et al., 2010). This is the gap the incubator in this project is intended to address: a stable, sealed, environmentally controlled stage-top system that can hold imaging conditions over long, continuous runs, making it possible to observe microcolony formation directly.

2.2 Live-Cell Imaging Environmental Demands

2.2.1 General environmental requirements

Live-cell microscopy places strict requirements on the sample environment. To avoid triggering stress responses that would alter the very processes being observed, cells must be kept at a stable temperature, with a controlled pH and minimal evaporation of the surrounding medium (Frigault et al., 2009). For mammalian cell culture, this pH control is usually maintained with a bicarbonate buffer and a 5% CO_2 atmosphere, together with a relative humidity of around 90 to 95% to limit evaporation (Talebipour et al., 2024; Coutu and Schroeder, 2013). Standard cell incubators are built to hold these conditions but are generally incompatible with microscopy, which has driven a shift toward compact stage-top systems that deliver heat, humidity and gas directly to the sample on the microscope itself (Talebipour et al., 2024).

2.2.2 Carbon dioxide and humidity requirements for mycobacteria

Mycobacteria are not cultured the same way as mammalian cells. Both *M. smegmatis* and *M. tuberculosis* are grown at around 37°C, but *M. smegmatis* is non-fastidious: ATCC's handling instructions for the standard laboratory strain (mc^{2}155) list the required atmosphere simply as aerobic, with no added CO_2 (American Type Culture Collection, 2026). *M. tuberculosis* often needs a 5 to 10% CO_2 atmosphere, but mainly for primary isolation from clinical samples or when growing from a very small number of cells, since CO_2 substitutes for a biotin requirement in many clinical strains, a mechanism first identified by Schaefer et al. (1955) and still reflected in current culture media guidance (Hardy Diagnostics, 2020). Established laboratory strains grown to a reasonable density in liquid culture are commonly kept at 37°C in ordinary air instead, with a CO_2 incubator reserved for the mammalian host cells in infection experiments rather than for the bacteria itself (Nazarova and Russell, 2017). Humidity in mycobacterial culture is also usually managed indirectly, for example by loosening or tightening tube caps, rather than held at a fixed relative humidity as in mammalian cell culture (Hardy Diagnostics, 2020).

This suggests the incubator in this project does not need to copy mammalian-style CO_2 and humidity control by default: *M. smegmatis* should grow well in ordinary air, and humidity control here is mainly needed to stop the small imaging volume from evaporating, not to satisfy a biological requirement of the organism. The system should still be able to support a controlled CO_2 atmosphere, but the reason for that is worth stating precisely, because it is not the bacterium. The intended progression of this work is toward host-cell infection models, where mycobacteria are imaged interacting with mammalian tissue: precision-cut lung slices, cells cultured at a Matrigel interface, or conventional monolayers. In those experiments it is the host tissue, not the bacterium, that fails without a bicarbonate-buffered 5% CO_2 atmosphere and near-saturated humidity (Nazarova and Russell, 2017; Frigault et al., 2009). Retaining a CO_2 loop is therefore not over-engineering for *M. smegmatis*; it is what allows the same chamber to be used later for infection imaging without a redesign.

2.2.3 Temperature stability

Temperature stability matters beyond simply keeping cells alive. Grigorov et al. (2023) showed that shifting *M. smegmatis* from 37°C down to 15°C halted cell division for around 24 hours and altered the expression of nearly a quarter of the genome within two hours, with growth-related genes downregulated and stress-response genes upregulated. A substantial deviation from the target temperature therefore does not just slow growth, it actively diverts the cell's resources toward survival, which risks confounding the growth and division behaviour the incubator is meant to observe. This also suggests a useful incubator should support an adjustable setpoint rather than only a single fixed temperature, since researchers may want to run temperature-shift experiments deliberately (Grigorov et al., 2023). There are two general approaches to holding temperature during imaging: enclosing the whole microscope in a heated box, which is stable but slow to settle and awkward to modify, or heating a small stage-top chamber around the sample, which is faster and more flexible but must be designed carefully, since immersion objectives act as a heat sink and can pull the sample below the target temperature at the exact point being imaged (Frigault et al., 2009).

2.2.4 Evaporation and humidity

Small sample volumes are also vulnerable to evaporation, which changes the osmolarity of the medium and can stress the cells (Frigault et al., 2009). The standard way to limit this is to humidify the air around the sample, and evidence from microscale cell assays suggests a minimum relative humidity of around 95% is needed, since below this level even small amounts of evaporation can generate local currents strong enough to disturb the sample, an effect that worsens as chamber volume shrinks (Berthier et al., 2008a,b). A simple way to help maintain this is to include a reservoir of water inside the sealed chamber, which evaporates in place of the sample and buffers against small leaks in the seal (Berthier et al., 2008b).

2.2.5 Focal-plane drift

Focal-plane drift is a further practical constraint on long-duration imaging. Most drift is caused by small changes in temperature rather than mechanical movement, and even good laboratory air conditioning is reported to leave a residual fluctuation of around 0.5°C, typically worse under the variable heat loads of a real lab (Adler and Pagakis, 2003). Drift is driven less by the absolute temperature than by how quickly it changes, since different parts of a microscope have different thermal mass and settle at different rates (Adler and Pagakis, 2003). Left uncorrected, drift can exceed the resolution of the microscope and distort three-dimensional image stacks, and in time-lapse data it can be difficult to distinguish from real cell movement or growth (Adler and Pagakis, 2003). Notably, localised heat sources such as sample chamber heaters and objective heaters, the same components needed to hold the sample at a stable temperature, are also identified as contributors to drift (Adler and Pagakis, 2003). This is a direct design tension for a stage-top incubator: the heater needed to stabilise temperature at the sample can itself become a source of focal drift, which is why characterising drift in the finished system is an important part of this project.

2.3 Existing Stage-Top Incubator Systems

A number of commercial and open-source stage-top incubators already exist. The commercial systems (Ibidi, Okolab, Tokai Hit and LCI) are built to a high standard but are closed, proprietary and expensive. They are, in principle, biology-agnostic environmental boxes: nothing about a heated chamber holding 37°C at a controlled humidity prevents it from hosting a bacterial sample. What they are is calibrated around mammalian cell culture, in their default setpoints (5% CO_2, near-saturated humidity), in the sample vessels they accept, and in the price point the mammalian cell-biology market will bear. The open-source and DIY systems, built mostly by academic groups (Talebipour, Worcester, Ragazzini, Yan and ThermoCyte), reproduce the same core functions at a fraction of the cost, usually with an Arduino or similar microcontroller. This section works through each environmental variable in turn, rather than each system in turn, so that the commercial and DIY approaches to the same problem sit side by side. Two points are common to nearly all of these systems and are worth stating once rather than repeating throughout: bulkiness is a shared drawback, a full chamber can be awkward to handle on a crowded stage, and some custom designs place heating elements directly next to the cell chamber, which risks electrical or magnetic interference with the experiment (Talebipour et al., 2024); and a chamber does not need to be fully airtight to work well, since reducing leakage mainly lowers the CO_2 needed to hold a setpoint, and so the running cost, rather than being required for stability (Talebipour et al., 2024).

2.3.1 Temperature control

Three basic approaches to temperature control appear across the systems reviewed: continuous or proportional heating below saturation, used by the commercial Ibidi and Okolab systems; full closed-loop PID, used by the commercial LCI system and the DIY builds from Ragazzini, Yan and ThermoCyte; and simple bang-bang switching, used by Worcester's DIY build (ibidi GmbH, 2026; Okolab s.r.l., 2026; Live Cell Instrument, 2021; Ragazzini et al., 2019; Yan et al., 2024; O'Carroll et al., 2023; Worcester et al., 2025). PID gives the tightest, most stable holding, typically within a few tenths of a degree, at the cost of more complex electronics and tuning effort. Bang-bang is far simpler to build and program but accepts a wider, oscillating error band, and a heat-only PID design that cannot actively cool, as in Ragazzini's build, will always overshoot slightly above setpoint rather than settling on it exactly.

How the heat is delivered also has optical consequences, which is easy to overlook when comparing systems on holding accuracy alone. Ragazzini et al. (2019) heat their chamber through indium tin oxide (ITO) coated glass, but cut a hole in the ITO layer and fit a plain glass coverslip in its place, specifically to restore the optical transparency that phase-contrast imaging requires. This detail matters beyond that one build: any heated window sits directly in the optical path, and reduced transmission through it has to be compensated by raising illumination intensity at the source, which is the standard driver of phototoxicity and photobleaching in long time-lapse runs (Frigault et al., 2009). Phototoxicity is not something the incubator controls, then, but the transparency of whatever material carries the heating element is a design choice that indirectly sets how hard the illumination has to work over a multi-day run.

Nearly every commercial system also heats the objective lens directly, whether as a built-in PID channel (LCI) or an add-on accessory (Tokai Hit, Ibidi). This is not a nice-to-have: Section 2.2.3 established that an unheated immersion objective acts as a heat sink and pulls the sample below its setpoint at the exact point being imaged. Ibidi's chamber design, shown in Figure 2.1, applies the same logic to the whole enclosure: the lid is held a few degrees above the plate, which creates a gradient that stops condensation forming on the optical path, while a separate humidity sensor and a pre-humidified gas mixture hold the chamber near saturation (ibidi GmbH, 2026).

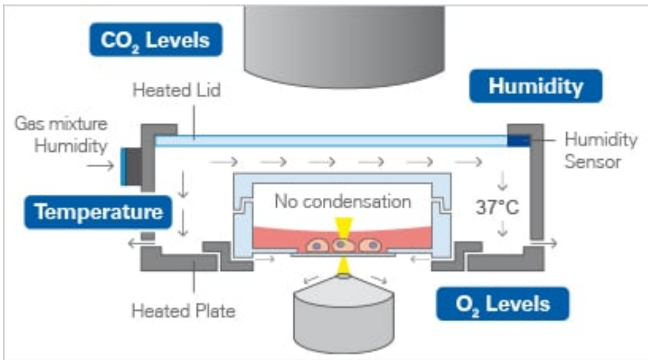

Figure 2.1: Ibidi stage-top incubator temperature, humidity and gas control schematic. The heated lid is held above the heated plate to prevent condensation forming on the optical path, while a humidity sensor and pre-humidified gas mixture hold the chamber near saturation. Reproduced from ibidi GmbH (2026).

Given that *M. smegmatis* tolerates a wider temperature range than the precision most of these systems target (Section 2.2.3), the real design question here is not which method is most accurate in absolute terms, but which is simple enough to build on the STM32 platform already specified while still holding within whatever tolerance the biology needs. A basic PID loop is the more defensible default: Ragazzini and Yan both show it is achievable for well under EUR 300 in DIY components, it can be reused across all three environmental loops instead of mixing control schemes, and it avoids the standing overshoot that comes with a heat-only bang-bang design. An objective heater is worth building in from the outset for the reason above, rather than retrofitting it once drift becomes a measured problem.

2.3.2 Carbon dioxide control

Two ways of delivering CO_2 appear across the systems reviewed: a small pressurised cylinder or capsule feeding a solenoid valve under sensor feedback, used by the DIY builds from Talebipour and Ragazzini and, commercially, as one of Tokai Hit's two cylinder options; and a larger premixed or in-line gas-mixing system, used by Yan's DIY build and the commercial LCI and Okolab systems (Talebipour et al., 2024; Ragazzini et al., 2019; Tokai Hit, 2026; Yan et al., 2024; Live Cell Instrument, 2021; Okolab s.r.l., 2026). Every system that controls CO_2 at all closes the loop with some form of NDIR sensor and a valve, regardless of which delivery route it uses; ThermoCyte is the one system reviewed with no CO_2 control, since it is built for a flow-through setup where gas concentration is not the limiting variable (O'Carroll et al., 2023). Unlike humidity (below), no system leaves CO_2 to hold its own level passively, since it is continuously consumed and lost to leakage rather than settling toward an equilibrium on its own.

The cylinder-and-solenoid route is the more defensible choice for this project. It is what the cheapest working DIY builds already use (Ragazzini sourced CO_2 from an aquarium supply rather than a laboratory gas line, correcting the setpoint from 5% to 5.3% to account for dilution by water vapour once the dry, measured gas mixes with the humidified chamber air), it needs no lab gas line, and it reuses the same sensor-and-valve logic, and potentially the same PID loop, as temperature control. Since *M. smegmatis* does not require added CO_2 (Section 2.2.2), the actual requirement here is narrower than matching the 5% target every reviewed system is built around. The loop needs to exist, and to be accurate enough to be tuned up later, because the experiments this platform is meant to enable next are host-cell infection models: mycobacteria imaged against precision-cut lung slices, Matrigel interfaces or mammalian monolayers, where the host tissue will not stay viable without a bicarbonate-buffered 5% CO_2 atmosphere (Section 2.2.2). Designing the CO_2 channel to mammalian-grade accuracy from day one is not required for the *M. smegmatis* demonstration, but designing it out entirely would foreclose the intended next step.

2.3.3 Humidity control

Two real options emerge here: an actively controlled external humidifier that blends humidified and dry gas before it reaches the chamber, used by the commercial Ibidi, Okolab and LCI systems and Tokai Hit's optional accessory; and a passive water reservoir inside the sealed, heated chamber that simply evaporates in place, used by the DIY builds from Talebipour, Ragazzini and Yan (ibidi GmbH, 2026; Okolab s.r.l., 2026; Live Cell Instrument, 2021; Tokai Hit, 2026; Talebipour et al., 2024; Ragazzini et al., 2019; Yan et al., 2024). The passive route is attractive: it needs no extra sensor or control loop, and three of the five DIY systems reviewed rely on it successfully even while running PID on temperature and CO_2. It is also the source of the clearest practical failure mode in this comparison, though the problem is reservoir capacity rather than the passive mechanism itself: a small reservoir runs dry on a timescale of hours to a few days, as Worcester's mint-tin ultrasonic atomiser (45–50 mL) and Tokai Hit's default internal reservoir (refilled every three to four days) both show, which does not fit the multi-day runs this project needs.

The fix, in other words, is reservoir size and access, not closed-loop control. Okolab's external module, Tokai Hit's optional external humidifier accessory and LCI's wider Incubator TW tank all solve this the same way, by moving the reservoir outside the chamber or simply making it bigger, so it can be topped up without disturbing the sample. A passive reservoir, sized and positioned so it can be refilled without opening the chamber, is therefore the more proportionate choice here than active gas humidification: it avoids adding a fourth feedback loop to an already coupled multivariable system, provided its capacity is designed around the run length this project actually needs rather than copied from a system built for shorter mammalian-culture imaging sessions.

2.3.4 Focal-plane drift

None of the systems reviewed here report a direct measurement of focal-plane drift, which is worth noting rather than glossing over. Section 2.2.5 established that drift is driven less by absolute temperature than by how quickly it changes, and that different microscope components, having different thermal mass, settle at different rates when disturbed. An unheated objective is a clear case of this: with no active control of its own, its temperature tracks the room rather than the sample, so an ambient swing in the lab pulls the objective and the actively stabilised sample stage apart at different rates, and that mismatch shows up as drift. Heating the objective to the sample's setpoint, as LCI, Tokai Hit and Ibidi all do, should reduce this specific driver: it brings the objective under the same active control as the rest of the chamber and largely decouples its temperature from ambient swings, rather than leaving it exposed to them. This is the same heat-sink logic established in Section 2.2.3, just applied to drift rather than mean temperature.

The same fix can cut the other way, though. Section 2.2.5 also identifies localised heat sources, including objective heaters themselves, as a contributor to drift, and a poorly regulated heater reintroduces exactly the kind of rapid, local temperature change that causes it. A bang-bang or heat-only PID objective heater, cycling the way Ragazzini's heat-only chamber heater does, would put a small, repeating temperature swing right next to the optics instead of removing one; a tightly tuned PID loop would not. So an objective heater is not simply good or bad for drift, whether it helps or hurts depends on how well it is itself regulated, and since no system reviewed here reports having measured the net effect, characterising drift in the finished incubator, with the objective heater on a well-tuned PID loop versus running bang-bang or open-loop, is both good practice and a genuine gap in how these systems are normally reported.

2.3.5 Interface and monitoring

Two options appear here as well: a dedicated display or touchscreen, in some cases with remote or mobile monitoring, used by all four commercial systems (LCI's Android app and alarm function being the most complete example) (Live Cell Instrument, 2021; Okolab s.r.l., 2026; Tokai Hit, 2026); or a bare-bones host-computer or Arduino-LCD setup, used by every DIY system reviewed, from Worcester's Visual Studio executable and Ragazzini's and Yan's LabVIEW logging to ThermoCyte's headless Arduino logging with no graphical interface at all (Worcester et al., 2025; Ragazzini et al., 2019; Yan et al., 2024; O'Carroll et al., 2023).

Given the STM32 platform and host-side logging already specified for this project, a simple serial or USB link to a logging computer, the approach Worcester, Ragazzini and Yan all used, is proportionate: sufficient for a lab prototype without spending build effort replicating LCI's remote app, which is worth revisiting later if it is not too costly to add.

2.3.6 Accuracy

Yan et al. (2024) directly benchmarked their own microscope incubation system (MIS) against three of the commercial systems reviewed above, on dimensions, temperature accuracy and gas accuracy, reproduced in Table 2.1. Their own device, at ±0.19°C and ±0.13% CO_2 by volume, sat within the same range as the commercial systems on temperature and comfortably within range on gas accuracy, despite being built at a fraction of the cost.

Table 2.1: Dimensions and control accuracy of three commercial stage-top incubators compared against the custom-built microscope incubation system (MIS). Reproduced from Yan et al. (2024).

	Okolab	Tokai Hit	Ibidi	MIS
Dimensions (mm)	160 × 110 × 29	187 × 146 × 27	127.5 × 85.5 × 25	160 × 130 × 29
Temperature accuracy	±0.1/±0.3°C	±0.3°C	±0.2°C	±0.19°C
Gas accuracy	±0.1% vol	±0.1% vol	±0.5% vol	±0.13% vol

A caveat applies to how these figures should be weighted against each other. Yan's numbers come from a single study that measured all four systems on the same bench under the same protocol, which is what makes the comparison meaningful; the commercial specifications quoted elsewhere in this section, including the Ibidi heat-up data below, are manufacturer-reported and have not been independently verified, and vendors do not generally publish the test conditions behind them. The commercial figures are therefore best read as claims rather than measurements.

Ibidi's own reported heat-up data, for comparison, show the sample reaching 37.0 ± 0.2°C within 40 minutes and holding that value even as room temperature is deliberately varied (ibidi GmbH, 2026). Among the DIY systems, Talebipour et al. (2024) report a 2°C differential between the heated supply water (39°C) and the cell culture wells, which settle at 37°C ±0.1°C. Ragazzini et al. (2019) held temperature at 37.0 ± 0.3°C. O'Carroll et al. (2023) achieved a mean temperature of 36.56°C with a standard deviation of 0.42°C over a one-hour hold, alongside a real thermal gradient of around 2°C across the dish. Worcester et al. (2025) held 90.8 ± 0.9% relative humidity, 36.8 ± 0.7°C and 5.3 ± 0.2% CO_2 over an eight-hour test run alongside a commercial Ibidi system for comparison, with comparable cell doubling times measured in both. Taken together, a carefully built DIY system, particularly one using PID control, reaches accuracy broadly comparable to commercial systems, even if the commercial systems remain more consistent overall.

One further limitation cuts across all of these figures: the hold durations reported are short. ThermoCyte's characterisation covers one hour, Worcester's eight, and none of the systems reviewed report a continuous multi-day characterisation of the kind a mycobacterial microcolony experiment would demand. Whether these accuracies survive over 48 hours or longer, once reservoirs deplete, sensors drift and the lab's own day–night ambient cycle is superimposed on the chamber, is simply not established in this literature.

2.3.7 Comparison and costs

Table 2.2 draws the cost, modifiability and per-variable control method of every system covered in this section into one place. Three points follow directly from it. First, PID control is the practical standard wherever precision matters on temperature or CO_2 (the commercial LCI system and the DIY builds from Ragazzini, ThermoCyte and Yan all use it), while bang-bang control, as used by Worcester's DIY build, is simpler and cheaper but accepts a wider, oscillating error band; humidity breaks this pattern, since Talebipour, Ragazzini and Yan (all DIY) leave it to passive evaporation even while actively controlling temperature and CO_2. Second, an objective heater is close to a standard feature across the commercial systems, for the heat-sink reason established in Section 2.2.3, so it is worth building in from the outset rather than adding later.

Third, and most directly relevant here, none of these systems were developed with mycobacterial imaging in view. That is a statement about their design targets, not their capability: a chamber that holds 37°C at controlled humidity will host *M. smegmatis* perfectly well, and Worcester's build was in fact validated against a commercial Ibidi system on mammalian doubling times rather than on any property specific to the cell type. The gap is elsewhere. The commercial systems are priced for well-funded cell-biology laboratories and are closed to modification, so a research group cannot add a sensor, change a setpoint range or adapt the chamber geometry to a different sample vessel. The DIY builds are modifiable and cheap but are individually documented one-off builds rather than a characterised platform. And across both categories, the environmental envelope is set by mammalian defaults, 5% CO_2 and near-saturated humidity held to tight tolerance, which is more control than *M. smegmatis* needs on CO_2 and says nothing about whether the system holds up over the multi-day runs microcolony formation requires. A compact, modifiable incubator, scoped to what mycobacterial imaging actually demands and characterised over realistic run lengths, is the gap this project is intended to fill.

Table 2.2: Comparison of stage-top incubator systems reviewed in this section. Approximate cost, modifiability, and the control method used for temperature, CO_2 and humidity in each system.

System	Approx. cost	Modifiable	Control method		
			Temperature	*CO_2*	*Humidity*
Ibidi Silver Line	R 422,600 (USD 26,085, full config.)	No (proprietary)	Continuous current, closed-loop	Optional CO_2/O_2 module	Active gas humidification, closed-loop
Okolab BL3	Not published	No (proprietary)	Active feedback, closed-loop	NDIR sensor, closed-loop	External module, 50–95% RH, closed-loop
Tokai Hit	Not published	No (proprietary)	Radiant heat, sample-fed-back	Cylinder-fed, 5–20%	Internal reservoir (external accessory available)
LCI Incubator T	Not published	No (proprietary)	PID, all four channels	PID, NDIR + solenoid valve	PID, heater/ sensor in reservoir
Talebipour mini-incubator	Not reported	Yes (open build)	Closed-loop, water jacket	Solenoid valve, 5%	Passive: humidifies incoming CO_2
Worcester DIYncubator	R 1,940–6,480 (USD 120–400)	Yes (open build)	Bang-bang	Bang-bang	Ultrasonic atomiser, small reservoir
Ragazzini incubator	Approx. R 4,860 (EUR 260)	Yes (open build)	PID, heat-only	PID, NDIR + solenoid valve	Two heated reservoirs, passive evaporation
Yan MIS	Not reported	Yes (custom build)	Incremental PID	Incremental PID, 5% ± 0.13%	Passive evaporation from reservoir
ThermoCyte	Approx. R 3,740 (EUR 200, electrical/ heating only)	Yes (open-source)	PID	None	None

Rand figures are approximate, converted at USD/ZAR 16.20 and EUR/ZAR 18.70 (rates as of August 2026); original currency shown in parentheses.

For context on commercial pricing, an undergraduate biomedical design report comparing available commercial options found a World Precision Instruments Stagetop Environmental Chamber with Controller priced at USD 11,500 and an Ibidi Stage Top Incubation system at USD 13,990 (Puccinelli), broadly consistent with the current Ibidi figure above, allowing for the fact that exact prices vary with configuration and change over time. The same report also compared chassis materials for a build of this kind, scoring polypropylene (PP) highest overall: it is the cheapest of the materials considered, autoclavable, and moulds easily, at some cost in heat insulation relative to HDPE (Puccinelli). This source is an undergraduate design report rather than peer-reviewed work, so it is used here only for the market survey and materials comparison it documents, not as an independent technical result. Autoclavability is not a requirement for the *M. smegmatis* prototype built in this project, but it is worth keeping in mind when choosing a chassis material, given this project's stated role as a stepping stone toward sterility- and containment-sensitive *M. tuberculosis* work.

2.3.8 Microfluidic alternatives for bacterial imaging

Stage-top incubators are not the only route to long-duration bacterial imaging. A parallel line of work builds microfluidic culture devices that trap individual bacteria in defined channels and perfuse fresh medium past them, and at least one such device has been developed specifically for *M. smegmatis* (Wang et al., 2021), with earlier mycobacterial systems reported by Golchin et al. (2012). These are the closest prior art to this project by organism, and it is worth being precise about why they do not close the same gap.

Two points stand out. First, these devices do not replace an incubator, they sit inside one: Wang et al. (2021) operate their chip within the incubation enclosure of the microscope, which handles temperature exactly as the systems in Section 2.3 do. Microfluidics solves medium delivery and cell positioning, not environmental control, so the control problem this project addresses remains unsolved by that route. Second, and more fundamentally, the design intent runs opposite to the biology of interest here. Wang et al. (2021) confine cells to chambers roughly 0.9 µm in height precisely to force growth into a single two-dimensional layer, because mycobacteria otherwise form overlapping three-dimensional clusters that are difficult to resolve optically. Their own framing notes that most microfluidic trapping devices were developed for mammalian cells and translate poorly to bacteria, which are smaller and non-spherical. That confinement is the right answer for tracking individual lineages, but it engineers away the multicellular organisation, cording and microcolony architecture, that this project sets out to observe. Reported culture durations are also on the order of 48 hours, short relative to the runs a slow-growing mycobacterial experiment would eventually require.

The conclusion is that microfluidic single-cell platforms and stage-top incubators answer different questions rather than competing. Neither offers a compact, modifiable, environmentally controlled chamber in which a mycobacterial microcolony can be left to form and be imaged as it does so, which is what this project is aimed at.

2.4 Multivariable Control of Coupled Environmental Loops

Section 2.3 surveyed how existing incubators generate and sense temperature, CO_2 and humidity, largely as three separate problems: a heated window or lid to stop condensation, a reservoir or humidifier to hold humidity near saturation, a solenoid valve to meter CO_2 into a small sealed volume. What that section did not address is what happens when all three are actuated at once in the same enclosure, which is the more general control problem this project actually has to solve. A wider literature on coupled temperature, humidity and CO_2 control, drawn here from bioprinting, industrial climate testing and greenhouse engineering rather than cell culture specifically, gives a clearer picture of how difficult that coupling is and how it is normally handled.

2.4.1 Why temperature, humidity and CO_2 are coupled

Matamoros et al. (2020) built a decoupled PID controller for a bioprinter atmospheric enclosure and, citing Yinping and Lu (2017) among others, note that temperature and humidity are "highly related variables, so their simultaneous control is complex and requires fine tuning." The coupling is not a minor effect: relative humidity was found to drop by 2 to 3% for every 1°C rise in temperature, and without an active controller running, both variables were observed to fluctuate over a wide range together rather than settling independently. CO_2 adds a further layer of cross-coupling on top of this, and Matamoros et al. found it significant enough that they excluded CO_2 from their PID loop entirely, injecting it open-loop from a pressurised bottle through a solenoid valve rather than closing a third feedback loop around it (Matamoros et al., 2020).

This is a direct warning for a project like this one, where temperature, humidity and CO_2 are all meant to be actively and simultaneously held in a small sealed chamber: actuating one variable will disturb the others, and the difficulty of that interaction should not be underestimated on the basis of the single-loop PID descriptions surveyed in Section 2.3, none of which report having characterised cross-coupling directly.

2.4.2 Modelling coupled enclosures

A recurring approach in this literature is to model the enclosure as a lumped heat balance and a lumped water-vapour mass balance, both non-linear "in minus out" differential equations of the general form used throughout temperature and humidity control processes (Buonomano et al., 2017; Kapen et al., 2019). Matamoros et al. (2020) write these for their enclosure as

$$C_v V \frac{dT_h}{dt} = Q_{in} - Q_{out} = (G_a \rho C_p T_c + Q_n) - \left(G_a \rho C_p T_h + \frac{T_h - T_o}{R} \right) \tag{2.1}$$

$$\rho V \frac{d(d_n)}{dt} = G_a \rho d_c + D_n - G_a \rho d_n \tag{2.2}$$

where T_h and d_n are the enclosure's inner temperature and humidity, T_c and d_c are the temperature and humidity of the air supplied to the chamber, G_a is the air mass flow rate, Q_n and D_n are the heat and moisture added by the actuators, R is the enclosure's thermal resistance, and ρ and C_p are the density and specific heat of air (Matamoros et al., 2020). Applying the Laplace transform to Equations 2.1 and 2.2 gives first-order transfer functions for temperature and humidity, $G_1(s) = T_h(s)/T_c(s)$ and $G_2(s) = d_n(s)/d_c(s)$ respectively, which is the standard route from a physical balance equation to a form a PID controller can be designed around (Matamoros et al., 2020). Riahi et al. (2020) apply essentially the same balance-equation structure to a much larger and leakier enclosure, an agricultural greenhouse, and validate the resulting model against several days of real sensor data rather than simulation alone; that the same modelling approach holds across enclosures this different in scale is a reasonable indication it would generalise down to the much smaller chamber in this project.

Two practical details from this modelling process are worth carrying forward. First, Matamoros et al. (2020) found their experimental and theoretical transfer functions diverged slightly until an exponential dead-time term was added to the proportional constant, attributed to sensor response lag rather than the physical system itself; any sensor used in this project should be characterised for this kind of lag before poor control performance is blamed on the plant instead. Second, both the theoretical and experimental transfer functions were fitted and validated against real MIMO/SISO test data using MATLAB's System Identification Toolbox, with an initial Ziegler–Nichols tuning subsequently hand-corrected to remove overshoot: a practical, tool-supported route to a working PID controller that does not require every physical constant in Equations 2.1 and 2.2 to be known in advance.

2.4.3 Coupled versus decoupled control architectures

There is no settled answer in this literature to whether temperature and humidity should be controlled by one coupled controller or by two independent loops. Matamoros et al. (2020) note explicitly that some prior work proposes coupled controllers while other work, including their own, proposes decoupled ones, and choose the decoupled route deliberately to keep the PID design and tuning tractable. Riahi et al. (2020) make the same architectural choice for a fuzzy rather than PID controller, separating temperature error and humidity error into two independent control loops rather than a single multi-input controller; the decision to decouple holds across both algorithm types, even though the underlying control law differs.

For a project with three coupled loops, temperature, CO_2 and humidity, rather than two, and a single embedded STM32 platform to implement them on, this is a reasonable argument for following the same decoupled pattern already found throughout Section 2.3, where nearly every DIY and commercial incubator runs one control loop per variable rather than a single coupled multivariable controller. A coupled controller may ultimately give tighter performance, but it adds design and tuning complexity that this literature does not treat as a settled requirement, only as one of two competing approaches.

2.4.4 Beyond PID

PID is not the only option for holding a coupled multivariable system at setpoint, though it is the one used throughout Section 2.3 and by Matamoros et al. (2020). Chen et al. (2025), reviewing environmental control strategies for agricultural greenhouses, a domain with the same coupled temperature, humidity and CO_2 problem at a much larger scale, place PID at the start of a developmental progression that also includes fuzzy, feedback, adaptive, model-predictive and neural-network control, with hybrid schemes combining several of these. Table 2.3, reproduced directly from that review, summarises the advantage, disadvantage and typical application of each.

Table 2.3: Comparison of control algorithms used for coupled environmental control. Reproduced from Chen et al. (2025); content and wording are the original authors', reformatted here from numbered sub-points into semicolon-separated lists to fit a table cell, not produced for this review.

S/N	Control algorithm	Advantage	Disadvantage	Application scenarios
1	PID control	Simple controller structure	Difficult to control multiple variables at the same time	Individual parameter control; primary greenhouse control
2	Fuzzy control	No need for accurate modelling	Poor real-time performance; constrained by the rule base	Multivariate greenhouse system; control of nonlinear systems
3	Feedback control	Highly accurate; predictable response behaviour	High sensor accuracy requirements	Scenarios for precise environmental control; real-time control requirements
4	Adaptive control	High system adaptive control capability; easy self-optimisation	Initial-phase system instability; difficulty in system development and maintenance	Self-adjustment of control strategies in response to environmental changes
5	Model predictive control	Predicting the future dynamic behaviour of the system; multi-constraint optimisation capability	Computational complexity; high model dependency	Multivariate systems control; complex system dynamics
6	Neural network control	Good self-learning skills; ability to process time-series data	High data demand; high training time and computational resource requirements	Scenarios combining prediction and control; systems requiring adaptive capabilities
7	Hybrid control	Integrating multiple control methods; high control accuracy and response speed	Complex design and commissioning; high real-time requirements; difficult and costly to realise	Multivariable and uncertain systems; systems requiring adaptive capabilities; scenarios combining prediction and control

Chen et al.'s own conclusion is that PID remains "the simplest and most widely used method," but is "no longer sufficient" as control demands grow, and that model-predictive and neural-network control are the current focus of active research (Chen et al., 2025). For this project, though, that growing sophistication is not obviously worth adopting: PID is the only one of these seven algorithms used anywhere in Section 2.3, it is the only one with a demonstrated low-cost, single-microcontroller implementation in the incubator literature specifically (Ragazzini, Yan, ThermoCyte), and the disadvantages of the more advanced alternatives, principally high model dependency for model-predictive control and high data and training demand for neural-network control, are a poor match for a system with a single prototype to tune on and no operating history to learn from. Yinping and Lu (2017) report an "expert control" algorithm for a similar heat-humidity test chamber, but do not report enough implementation detail to evaluate it as a serious alternative here. PID, run as decoupled per-variable loops for the reasons set out above, remains the more defensible choice for this project; the more advanced algorithms in Table 2.3 are worth revisiting only if PID's coupling-driven error proves too large to tolerate once the finished system is characterised.

2.5 Summary of the Literature

Taken together, the literature reviewed in this chapter leaves a specific and unfilled space. The biology sets the requirement: *M. smegmatis* needs a stable, adjustable temperature and protection from evaporation over runs long enough for a microcolony to form, but not the 5% CO_2 atmosphere mammalian culture is built around, though a CO_2 channel must remain available for the host-cell infection models this platform is intended to serve next (Section 2.2.2). The existing stage-top systems reviewed in Section 2.3 can hold those conditions, but the commercial ones are closed and priced beyond a project of this scale while the DIY ones are individual builds rather than characterised platforms, and none of them report performance over the multi-day durations this application needs. The microfluidic devices that come closest by organism do not solve environmental control at all, and deliberately suppress the multicellular growth this project exists to observe (Section 2.3.8). The control literature in Section 2.4 establishes that the three environmental variables are genuinely cross-coupled and that decoupled per-variable PID is the defensible starting architecture, yet no incubator in Section 2.3 reports having characterised that coupling in a chamber of this size. No single system, therefore, combines compact and modifiable hardware, an environmental envelope scoped to mycobacterial imaging while retaining CO_2 capability, decoupled multi-loop control implemented on one embedded platform, and characterised long-duration performance including focal drift. That combination is what this project sets out to build and demonstrate.

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